A Two-Channel Accretion Mechanism for Evolving Dark Energy

Christopher P. Green
Independent Researcher
research@wilson-green.com

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Abstract

We present a physical mechanism for the evolving dark-energy signal in recent combined DESI analyses. A single accretion flux onto a parent black hole — whose interior is the observable universe — drives two competing channels: a fraction κ of the inward energy converts to interior vacuum (injection, raising $\rho_{\rm de}$), while horizon growth dilutes it (dilution). Injection spreads over the interior volume ($\propto a^3$) and dominates early; dilution tracks the boundary and dominates late. The dark-energy density therefore rises, peaks, and declines, and $w_{\rm eff}$ crosses $w = -1$ from below — phantom past, quintessence present — with no phantom fields. This ordering is *not fitted*: for $\kappa \leq 1$ the model provably cannot place the phantom phase in the present. Fit against Planck distance priors, DESI DR1 BAO, and full Pantheon+, it attains $\chi^2 = 1403.3$ — identical to the empirical w_0–w_a parameterisation and $\Delta\chi^2 \simeq 6$ below ΛCDM (1409.4) — with the crossing at $z \simeq 0.40$ and efficiency $\kappa \simeq 0.85$, inside the budget. A six-model Bayesian comparison favours the mechanism over the w_0–w_a, wCDM, and running-vacuum alternatives and finds it indistinguishable from ΛCDM and thawing quintessence; against DESI DR2 the preference strengthens ($\Delta\chi^2 \simeq 10$, κ easing to 0.69) and *it becomes the evidence-preferred of the six.* The dilution channel is the Misner–Sharp horizon relation and the injection the Oppenheimer–Snyder matching; two of the three junction layers are now established, only the conversion step open. Finally, because the boundary is a temporal surface — unlike a pure equation-of-state fit — uneven or rotating accretion breaks global isotropy, a natural route to the large-scale CMB anomalies (§8, not fitted here). The model makes one central bet: the evolving-dark-energy preference ($\sim 2\sigma$ here, 2.8–4.2σ in DESI DR2) is real and has this shape — and fails with it if surveys converge on ΛCDM.

1 Introduction

Combined analyses of DESI baryon-acoustic-oscillation data with the cosmic microwave background and Type Ia supernovae currently prefer an evolving dark energy over a cosmological constant: at the 3.1σ level for DESI DR2 BAO combined with the CMB, rising to 2.8–4.2σ when supernovae are added, depending on the sample used [1, 2]. The completed five-year survey (more than 47 million galaxies and quasars, with observations extended to 2028) is under analysis, with first full-survey dark-energy results expected in 2027 [3]. The preferred w_0–w_a fits [4, 5] describe a dark energy that was phantom-like in the past ($w < -1$), crossed $w = -1$ near $z \sim 0.4$, and is quintessence-like today [2] — a history that is awkward to realise: fundamental phantom fields are widely regarded as pathological [6, 7], and models that cross $w = -1$ typically do so by construction rather than for a physical reason. The signal therefore sits between two readings: an artifact of the two-parameter form, or new physics with no accepted mechanism.

This paper supplies a candidate mechanism in which that specific history is forced rather than fitted, and reports its confrontation with data. The mechanism arises within a black-hole-interior cosmology: the observable universe is modelled as the interior of a black hole embedded in a parent spacetime and fed by ongoing accretion. That this feeding is continuous is the premise that sets the mechanism apart: earlier black-hole-interior cosmologies invoke the parent only at origin — through gravitational collapse or a bounce — whereas here it stays dynamically coupled to the interior today, and it is this sustained accretion across the horizon that drives the dark-energy history. Section 2 states the geometric premises. The mechanism's two channels are individually established physics — one is a geometric identity, the other a published conversion process — and the data confrontation of Sections 4–5 is independent of the geometric interpretation. Section 6 presents the supporting matching arguments, Section 7 the relation to prior work, Section 8 the consequences that follow from the geometric setting rather than from the fitted mechanism, and Section 9 the decisive tests.

Against Planck distance priors, DESI BAO, and the full Pantheon+ sample, the mechanism matches the empirical w_0–w_a description to within rounding (χ^2 of 1403.3 versus 1403.4, against 1409.4 for ΛCDM) — it captures the entire measured preference for evolving dark energy — while its crossing redshift, density peak, and present-day w emerge as outputs, not inputs, and its single bounded parameter, an energy-conversion efficiency, is selected by the data at $\kappa \simeq 0.85$, within the physical budget of $\kappa \leq 1$. A Bayesian model comparison sharpens the picture. Among six models fit on the same vector — ΛCDM, constant-w (wCDM), running-vacuum (RVM), thawing quintessence, the empirical w_0–w_a form, and the mechanism — the evidence favours the mechanism over all three dynamical alternatives (w_0–w_a, wCDM, and running vacuum, by $\Delta \ln \mathcal{Z} \simeq +2.5$ to $+2.7$) and leaves it statistically indistinguishable from ΛCDM and quintessence, disfavoured by none (Appendix D). It reproduces the evolving-dark-energy preference at the empirical form's fit quality while being *preferred* over that form despite the Occam penalty for prior volume being applied to it. This is where being a mechanism rather than a parameterisation tells: its efficiency is *physically bounded*, not freely fitted, so matching ΛCDM's evidence — rather than being penalised for three extra parameters — indicates those parameters are tracking genuine structure in the data, not buying fit through flexibility. Appendix D makes this concrete: the evidence advantage is tied to the physical $\kappa \leq 1$ bound and disappears exactly when that bound is lifted. On the newer DESI DR2 vector this strengthens: the fit improves to $\Delta\chi^2 \simeq 10$ below ΛCDM, the fitted efficiency eases from $\kappa \simeq 0.85$ to $\simeq 0.69$ (further inside the physical budget), and the mechanism becomes the evidence-preferred model of the six (Appendix E).

The proposal is falsifiable at both ends. If the strict junction analysis for a fed boundary overturns the volume accounting of Section 6, the mechanism fails theoretically. If the evolving-dark-energy signal dissolves as surveys mature, it fails empirically. It is stated precisely enough to lose.

2 Geometric setting

Four premises define the setting. The first two are familiar from the black-hole-cosmology literature; the third and fourth generate the mechanism.

(i) The universe as a black-hole interior. That the observable universe's mass and size satisfy a relation close to the Schwarzschild condition has been noted for decades [8], and interior cosmological solutions — collapse halting at extreme density and rebounding into an expanding interior — are an established line of work in general relativity and its torsion-extended variants [9, 10]. The "Big Bang" is a bounce inside a horizon, not a beginning from nothing.

(ii) Ordinary matter fixed early. Big-bang nucleosynthesis and the CMB tightly fix the early matter content [11, 12]. The interior baryonic and dark matter are established early and subsequently only dilute; the late-time channel below feeds the dark-energy sector only.

(iii) A one-way horizon delivering accretion inward. The interior's boundary is the parent's event horizon: nothing exits, and matter the parent accretes crosses inward. Inside a horizon the causal roles of the radial and time coordinates exchange, so infall is directed toward the interior's future. A fraction κ of the infalling rest energy converts, on arrival, into interior vacuum energy: the **injection** channel.

(iv) The same feeding grows the container. Accretion grows the parent's mass and horizon, enlarging the interior and diluting its vacuum: the **dilution** channel. With injection switched off ($\kappa = 0$) this limit can only weaken the dark energy, forcing $w \geq -1$ at all times; it serves below as the reduced model.

The essential content is that (iii) and (iv) are two channels driven by the same flux, with opposite signs and different volume dependence — injection into the interior's expanding volume ($\propto a^3$), dilution tied to the boundary. Their competition evolves: injection dominates while the universe is small, dilution as it grows. Dark energy strengthens, peaks, and weakens, in that order only.

3 The model

3.1 Background equations

For a spatially flat FRW interior with $a_0 = 1$ today:

$$H^2(a) = \frac{8\pi G}{3}\big[\rho_{\rm m}(a) + \rho_{\rm r}(a) + \rho_{\rm de}(a)\big], \tag{1}$$

with $\rho_{\rm m} \propto a^{-3}$, $\rho_{\rm r} \propto a^{-4}$. The new physics resides in $\rho_{\rm de}$.

3.2 Feeding history

With $M_p(a)$ the parent's mass, non-decreasing because feeding is one-way, the fractional feeding rate is parametrised as

$$\frac{d\ln M_p}{d\ln a} = \mu_0\, a^{\,n}, \qquad \mu_0 \geq 0, \tag{2}$$

giving $M_p(a) = M_p(1)\exp\big[-\mu_0(1-a^n)/n\big]$. The pair (μ_0, n) encodes the accretion history as seen from inside; Section 6 shows the same pair absorbs the parent-to-interior clock mapping.

3.3 The two channels

In units of today's dark-energy density, $\rho \equiv \rho_{\rm de}/\rho_{\rm de,0}$:

$$\boxed{\frac{d\rho}{d\ln a} = \mu_0\, a^{\,n}\Big[\underbrace{\kappa\,\frac{M_p(a)/M_p(1)}{a^3}}_{\text{injection }(+)} - \underbrace{2\,\rho}_{\text{dilution }(-)}\Big].} \tag{3}$$

The prefactor is the feeding rate: neither channel operates when the parent is not feeding. The injection term is the converted infall energy $\kappa\,\dot{M}_p c^2$ shared over the interior's physical volume, hence a^{-3}, with $\kappa \leq 1$ on energy grounds. The dilution coefficient is not adjustable: it follows from the horizon slaving $\rho \propto M^{-2}$ derived in Section 6, which this term reproduces exactly when $\kappa = 0$.

3.4 The forced ordering

The effective equation of state is $w_{\text{eff}} = -1 - \frac{1}{3}\, d\ln\rho/d\ln a$: phantom-like while injection dominates (density rising), quintessence-like while dilution dominates (density falling), with a crossing of $w = -1$ from below at the handover.

The ordering of the phases is a theorem of the model, not a fit outcome. At the present epoch, with $\rho(1) = 1$ and $M_p(a)/M_p(1) = 1$, the bracket evaluates to $\kappa - 2$: for any $\kappa < 2$ — and physically $\kappa \le 1$ — and any $\mu_0 > 0$, the density is declining *today* and $w_{\text{eff}}(0) > -1$ strictly. Any phantom phase is confined to the past, where the a^{-3} factor can lift the injection above the dilution. No parameter choice with $\mu_0 \ge 0$, $\kappa \le 1$ produces the reverse history (quintessence past, phantom present). The model can therefore fail against data that prefer the reverse ordering, and could never have been tuned to accommodate it.

The phantom phase involves no phantom matter. No component has superluminal or negative-kinetic-energy character; the apparent $w < -1$ is the standard signature of energy injection into the dark sector, familiar from interacting-dark-energy models [13, 14]. The one-way horizon — which for $\kappa = 0$ forbids $w < -1$, the wall only moving outward — is precisely what permits the phantom appearance: the flux is inward, an injection, and injection mimics phantom.

3.5 Energy conditions and the covariant completion

The model is effective rather than fundamental: it is defined by the sourced continuity equation of §3.3 (a physical energy-exchange law, not a fitted $w(z)$), but it is not derived from an action, and no explicit Lagrangian for the conversion vertex is claimed — a status shared with the published cosmologically-coupled-black-hole programme, whose microphysical realisation is likewise contested [15–17]. What can be established is the energy-condition budget of the minimal covariant completion, and that budget is clean.

Local conservation, enforced by the Einstein equations, forbids energy appearing in the dark sector ex nihilo: injected energy requires a physical carrier. The natural carrier is the infall itself — an ingoing null flux of Vaidya type [18], $T_{\mu\nu} = \varepsilon\, k_\mu k_\nu$ with k_μ null and $\varepsilon \ge 0$, crossing the horizon and converting to vacuum energy in the interior. Every component then satisfies the weak energy condition pointwise: the null dust by construction, the vacuum term $T_{\mu\nu} = -\rho_{\text{de}}\, g_{\mu\nu}$ because $\rho_{\text{de}} \ge 0$ (a positive density that the injection *raises*), matter and radiation trivially. The null energy condition is nowhere violated — the vacuum saturates it and every other component obeys it strictly — and the conversion is an internal transfer within a conserved total. The apparent $w < -1$ arises only in reconstruction: an observer who compresses vacuum, inflowing flux, and conversion into a single non-interacting fluid infers $w_{\text{eff}} = -1 - Q/(3H\rho_{\text{de}}) < -1$, the established super-acceleration signature of dark-sector energy exchange [13]. No local observer in any frame measures a negative energy density or an energy-condition-violating flux at any point in the interior domain.

Injection from the boundary also does not imprint a radial gradient. Inside a horizon the causal roles of the radial and time coordinates exchange, so the boundary is a past surface rather than a spatial wall: flux across it enters the interior spatially distributed and localised in time — exactly the structure of the feeding term $\mu_0 a^n$, homogeneous in space and dependent on epoch.

What remains open is the conversion vertex as an explicit healthy field theory — the absence of ghosts and superluminal modes in whatever field realises the transfer — an unbuilt construction shared as an open problem with the inside-out conversion programme; the flux's profile and normalisation belong to the junction analysis of §6.

The present treatment accordingly adopts the *rapid-conversion limit*: energy transferred across the accreting boundary is taken to convert into the dark-energy sector on timescales short compared with H^{-1}, so that the source may be treated as an effectively homogeneous component distributed over the FRW interior volume. Finite conversion times would introduce memory effects and are deferred; Appendix C states the assumption's scope, its minimal finite-Γ generalisation, and the grounds on which the fast limit is motivated.

4 Confrontation with data

Two data vectors are fit: a conservative one — Planck CMB distance priors (ℓ_A, R) [12, 19] plus five DESI BAO measurements (D_M/r_d, D_H/r_d at $z = 0.51$–2.33) [1] — and the full combination adding the complete Pantheon+ sample [20, 21]: 1580 SNe after standard $z > 0.01$ and calibrator cuts, full statistical+systematic covariance, analytically marginalised over the absolute-magnitude offset. Free parameters: Ω_m, H_0, (μ_0, n, κ). Baselines on the same vectors: ΛCDM and the w_0–w_a form [4, 5]. The results are collected in Table 1. Every fit and every Bayesian evidence value in the main text uses this same first-release (DR1) DESI BAO vector with Planck priors and Pantheon+; a refit of the whole analysis against the DESI second data release (DR2) is given as a robustness check in Appendix E, and DESI's DR2 field-level significance [2] is quoted below only as external context.

Table 1: Fit quality of the two-channel mechanism against the standard model (ΛCDM) and the empirical w_0–w_a parameterisation, on the conservative vector (Planck distance priors + DESI BAO) and on the full vector including the complete Pantheon+ sample. "Origin of $w(z)$" names what produces each model's dark-energy behaviour — a fixed value, a flexible fitting function, or a physical mechanism whose $w(z)$ is an output; "Free params" counts the free parameters varied in each fit.

Model	Origin of $w(z)$	χ^2 (CMB+BAO)	χ^2 (+1580 SNe)	Free params
ΛCDM	fixed value	17.6	1409.4	2
w_0–w_a (phantom allowed)	fitting function	12.9	1403.4	4
Two-channel mechanism	**accretion mechanism**	**13.8**	**1403.3**	5

With supernovae included, evolving dark energy is preferred over ΛCDM at $\Delta\chi^2 \simeq 6$, and the pipeline recovers the published best fit of comparable professional analyses ($w_0 \simeq -0.84$) [2]. The mechanism captures that entire preference at the empirical parameterisation's fit quality. Best-fit parameters on the full vector: $\mu_0 \simeq 0.37$, $n \simeq 1.8$, $\kappa \simeq 0.85$.

Shape. The best fit runs $w_\mathrm{eff} \simeq -1.14$ at $z = 1$, crosses -1 at $z \simeq 0.40$, and reaches $w_\mathrm{eff} \simeq -0.86$ today; the density peaks at $z \simeq 0.40$ (Figure 1). The data-side reconstruction independently shows a crossing near $z \simeq 0.4$, a peak near $z \simeq 0.4$–0.5, and $w_0 \simeq -0.84$. These are outputs of the fit; the ordering is fixed by Section 3.4 before any data enter. Model-agnostic reconstructions of current data independently describe an *emergent* dark energy — negligible at $z \gtrsim 1$, with the acceleration slowing at low redshift [22, 23] — which is the model's rise–peak–decline stated in reconstruction language.

Efficiency. Although bounded by $\kappa \leq 1$ on energy grounds, κ was fitted freely (allowed up to 40). The data select $\kappa \simeq 0.85$. On the conservative vector the $\chi^2(\kappa)$ profile, with the other parameters re-optimised, is a shallow valley bounded above by the ΛCDM value: $\mu_0 \to 0$ switches the mechanism off and recovers ΛCDM ($\chi^2 = 17.6$) at any κ, so the profile cannot exceed it. Converged values are $\chi^2 \simeq 16.2$ at $\kappa = 0.5$, 13.8 at $\kappa = 1$, and $\simeq 17.2$ by $\kappa = 3$, where the optimiser retreats to near-zero feeding. The free-κ fit continues past the physical ceiling into unphysical $\kappa > 1$, so the conservative optimum is quoted at the ceiling: $\chi^2 = 13.8$ at

$\kappa = 1$ (Table 1). Adding the supernova sample pins the selected value to $\kappa \simeq 0.85$, comfortably within the budget. Both lie within the profile's shallow minimum, and the mild shift reflects the supernovae's additional low-redshift leverage rather than any tension in the fit. The full improvement over ΛCDM is available only in the neighbourhood of the physical budget.

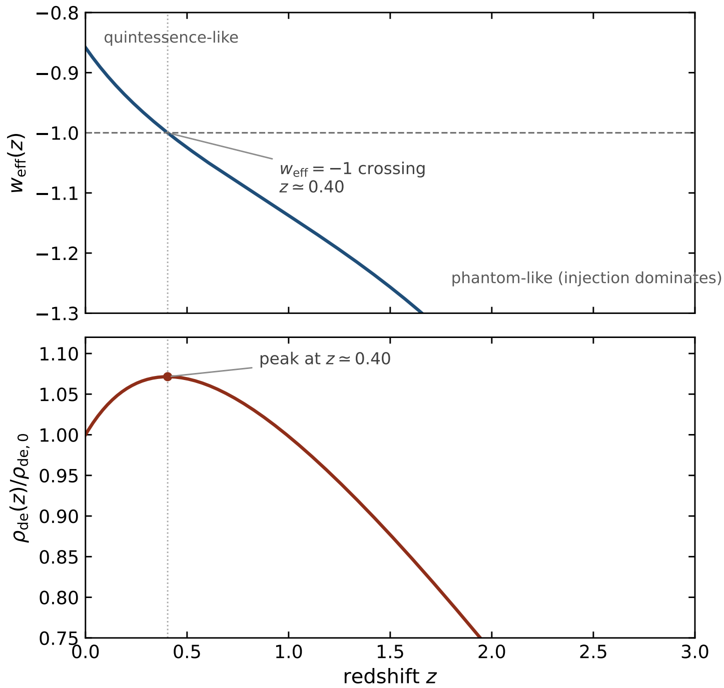

Figure 1: Best-fit history of the two-channel mechanism, obtained by integrating Eq. (3) at the best-fit parameters of this section ($\mu_0 = 0.37$, $n = 1.8$, $\kappa = 0.85$). *Top:* effective equation of state $w_{\rm eff}(z)$, crossing $w = -1$ from below at $z \simeq 0.40$. *Bottom:* dark-energy density in units of its present value, peaking at the crossing and declining thereafter; the crossing and the peak coincide by construction (Section 3.4).

Statistical assessment. On the fitted DR1 vector, $\Delta\chi^2 \simeq 6$ for three additional parameters is a $\sim 2\sigma$-level preference over ΛCDM, and the frequentist information criteria score the mechanism roughly level with ΛCDM on that same vector. Separately, and not from any fit performed here, the field-level evolving-dark-energy preference has since strengthened to 2.8–4.2σ in DESI's second data release, depending on the supernova sample [2], with the completed five-year survey now under analysis [3]. The mechanism reproduces the observed signal in shape and amplitude, leaves no significant residue for the empirical form to explain, and stands or falls with the signal as it matures.

Bayesian model selection. A six-model Bayesian evidence comparison on this vector — over which the mechanism is *favoured over every dynamical rival tested* (the empirical w_0–w_a form, wCDM, and running vacuum; $\Delta \ln \mathcal{Z} \simeq +2.5$ to $+2.7$) and *evidentially indistinguishable* from ΛCDM and thawing quintessence — is developed in full, with its cross-validation and a stress test that ties the advantage to the physical $\kappa \leq 1$ bound, in Appendix D. Its DESI DR2 counterpart, where the mechanism becomes the evidence-preferred model of the six, is in Appendix E.

Perturbation-level check. The best-fit $w_{\rm eff}(a)$ was tabulated and passed to the CAMB Boltzmann code [24] through its PPF module [25], which evolves perturbations stably across the $w = -1$ crossing, under the minimal completion of the interaction: homogeneous energy transfer and standard sound speed. The professional code reproduces this paper's background pipeline to 0.01% in the BAO observables and to 0.01 Mpc in the sound horizon. At fixed cosmological parameters the model's CMB TT spectrum differs from ΛCDM by at most +0.4% at $\ell = 2$–10 (late integrated Sachs–Wolfe, far beneath cosmic variance of tens of percent per multipole) and by 0.06% through the acoustic peaks; the growth observable $f\sigma_8(z)$ differs by at most 2% over $0 < z < 1.5$, against current measurement precision of 5–10%. The model therefore survives its first perturbation-level exposure, and neither the ISW tail nor structure growth discriminates it from ΛCDM or w_0–w_a at present precision: the discriminating data remain the expansion history itself.

5 Parameterisation artifacts

The w_0–w_a form can manufacture crossings. Taking the $\kappa = 0$ limit — whose equation of state provably never drops below -1 — at a data-allowed parameter point, generating mock observations, and fitting the mock with w_0–w_a recovers $w_0 = -0.97$, $w_a = -0.05$: an implied phantom crossing at $z \simeq 1.2$ that the underlying model does not contain.

This bears on the proposal in two ways. It softens the empirical case for a genuine crossing, since part of any reported crossing can be an artifact of the two-parameter form; and it implies that a genuinely crossing model cannot be distinguished from a thawing non-crossing one by the w_0–w_a numbers alone. The discriminating measurement is the full expansion history. Whether the DESI preference reflects a genuine phantom-divide crossing or is instead an artifact of the rigid two-parameter CPL form is itself debated [26, 27]; the present mechanism is less exposed to that ambiguity than a fitted crossing would be, since its rise–peak–decline ordering is forced by the theorem of Section 3.4 rather than chosen to match a particular empirical shape.

6 The matching problem

One physical choice carries the result: the injection shares its energy over the interior's physical volume ($\propto a^3$) rather than the parent's horizon volume. Under horizon-volume accounting the injection would scale with the boundary exactly as the dilution does, their ratio would freeze, and the crossing would vanish. Three results bear on the choice; one question remains open.

The dilution channel is an identity. The Misner–Sharp mass relation [28] $M = \frac{4\pi}{3}\rho\,(ar_b)^3$ combined with the boundary-at-horizon condition $ar_b = 2GM/c^2$ gives $\rho = 3c^6/32\pi G^3 M^2$: density falling exactly as M^{-2}. The dilution term reproduces this slaving numerically to one part in 10^5. It is the statement that the interior vacuum remains consistent with the horizon condition as the parent grows.

The two channels are independent mechanisms driven by one mass function. The injection and dilution terms are not two sides of a single conservation identity. Injection is energy transfer across the accreting boundary, set by the growth *rate*: by Eq. (2), $\mu_0 a^n M_p = dM_p/d\ln a$, so the injection term of Eq. (3) is $\kappa\,(dM_p/d\ln a)/a^3$ (Appendix A). Dilution follows independently from the horizon-slaving relation $\rho \propto M_p^{-2}$ (Appendix B) and, unlike ordinary FRW dilution, is gated by feeding. Both channels are legitimately driven by the same M_p: in spherical symmetry the Vaidya mass function coincides with the Misner–Sharp mass.

The exact matchings use interior-volume accounting. The Oppenheimer–Snyder construction [29] — the exact matching of an FRW interior to a black-hole exterior — identifies the exterior mass with interior density times the interior's physical FRW volume, the quantity scaling as a^3. Energy that has crossed the horizon belongs to the interior's budget and dilutes with the interior's expansion; horizon-volume accounting would import the boundary's geometry into the interior's local bookkeeping, which the exact solutions do not do.

The clock mapping is absorbed. The relation between parent proper time and interior cosmic time is strongly nonlinear near a horizon and is not derived here. Because the feeding history is fitted as a free function of the interior scale factor, any monotonic re-mapping of clocks is absorbed into (μ_0, n): the ambiguity affects only the translation of the fitted history into the parent's own schedule, not any prediction on our side of the horizon.

Open. Oppenheimer–Snyder matches a dust interior across a comoving, non-feeding boundary. An actively fed horizon is Vaidya-like [18] and the boundary advances as the parent grows; the strict junction analysis for that case [30, 31], and the perturbation-level treatment, remain to be done. The question is sharp: does active feeding modify the interior-volume accounting the exact static matchings support?

7 Relation to prior work: what is new and what is not

The present mechanism differs from previous black-hole-cosmology proposals — black-hole-interior models in general, holographic dark energy, and cosmologically coupled black holes — in that the dark-energy evolution arises from the competition between an accretion-driven injection term and an accretion-driven dilution term sourced by the same feeding history.

7.1 Existing ingredients

Black-hole-interior and bounce cosmologies constitute an established literature [8–10]. A horizon-set vacuum density ($\rho \propto R^{-2}$) is the defining relation of the holographic dark-energy programme [32, 33], which includes a published black-hole-universe realisation with a static horizon [34]. Matter-to-dark-energy conversion by black holes has been published for holes within our universe [15], including a 2024 result recovering the DESI time evolution from cosmologically coupled black holes [16]. Claimed direct observational support for that coupling exists — supermassive black holes in elliptical galaxies growing 8–20× beyond accretion, with zero coupling formally excluded at high confidence [35, 36] — and is contested by independent constraints from globular-cluster black holes [37], Gaia binaries [38], high-redshift AGN [39], and accretion histories [40], with the underlying theoretical argument also disputed [17]. The question is unresolved. Finally, an effective phantom equation of state produced by energy injection into the dark sector is standard interacting-dark-energy physics [13, 14].

7.2 Novel elements of the present mechanism

1. A single accretion flux sourcing *both* an injection and a dilution channel, with their relative scaling fixed by geometry (a^3 against the boundary) rather than by choice.

2. The consequent theorem (Section 3.4) that the rise–peak–decline history and the past-only phantom phase are forced: the reverse ordering is unattainable for $\kappa \leq 1$.

3. The outward-facing (parent-hole) realisation of conversion-as-dark-energy, in which the one-way horizon that forbids $w < -1$ in the dilution-only limit is what licenses the injection.

4. The demonstration that this mechanism captures the full DESI+Pantheon+ evolving-dark-energy preference at the empirical parameterisation's fit quality, with the fitted efficiency landing inside the physical budget.

5. A preferred direction that the same geometry naturally *permits* — because the boundary is a temporal surface, anisotropic feeding enters the interior as a large-scale directional modulation — offering a qualitative route to the observed large-angle CMB anomalies. This is permitted by the mechanism, not required by it: isotropic feeding yields the identical expansion history with no axis, and the amplitude is not yet derived (§8).

Taken together, these elements appear to be jointly novel, and the novelty is specific: it is not that the model is a mechanism — many candidates are — but that no published mechanism also does the other two things at once. A structured literature search (2023–2026) asked whether any single published model simultaneously (i) fits the data at least as well as w_0–w_aCDM, (ii) derives $w(z)$ from a genuine physical mechanism rather than a parametrisation, and (iii) naturally accommodates a large-angle CMB directional anomaly. The fit-plus-mechanism candidates are all isotropic and silent on the low-quadrupole and axis anomalies: interacting or coupled dark energy crosses $w = -1$ by dark-sector energy exchange with no associated anisotropy [13, 41]; the cosmologically-coupled-black-hole programme — the closest relative, recovering the DESI evolution from a mechanism with fewer parameters — is likewise isotropic, with its base version giving constant $w = -1$ and its microphysics contested [16, 17]; and thawing or quintom quintessence matches w_0–w_a but is isotropic and, for canonical fields, does not truly cross the phantom divide [27]. Conversely, the analyses that *do* fit an anisotropic dark-energy signal reach it through a binned parametrisation of the stress rather than a derived mechanism [42]. No single published model was found to combine all three properties. The present mechanism does, with the axis a permitted (not required) consequence of anisotropic feeding (§8); the combination therefore appears novel, though a literature search cannot exclude prior work in venues or under terminology not covered.

8 Speculative extensions

The following consequences arise from the broader black-hole-interior framework rather than from the fitted two-channel mechanism itself and may, if necessary, be discounted without prejudice to Sections 3–7.

A large-scale cutoff. A universe bounded at the parent's scale cannot support fluctuations larger than the container. Implementing a hard primordial cutoff in the Boltzmann code quantifies the prediction and discriminates between geometric conventions. With the largest supported wavelength equal to $2\pi R_s$, the suppression is confined to the quadrupole and octopole (D_2 reduced to 79% of ΛCDM, D_3 to 95%, no effect beyond $\ell = 4$) — matching the location and shape, though only part of the amplitude, of the observed low-quadrupole anomaly [43, 44]. With the largest wavelength equal to the diameter $2R_s$, power is suppressed to $\sim 35\%$ through $\ell \simeq 5$ and 69% at $\ell = 7$: a broad shelf the data do not show, disfavouring that convention. Both variants recover ΛCDM by $\ell = 10$. One conditionality attaches: the container's comoving size, R_s/a, was vast early and is at its minimum today, so whether any cutoff imprints at all depends on whether modes must fit the instantaneous comoving cavity or only the container at their generation — a boundary-condition question belonging to the junction analysis of Section 6. Under the generation-epoch reading no cutoff arises.

A preferred axis. A rotating (Kerr) parent or uneven feeding would imprint a preferred direction, potentially connecting to reported large-angle CMB alignments [43, 45]. The premise that a rotating parent black hole imprints a preferred axis on its interior, with the associated weakening effect tied to the DESI dark-energy signal, was reached independently by Popławski [46] through a classical route — the centrifugal effect of inherited rotation diminishing as the interior expands. The present mechanism differs at the level of what produces the axis: rather than inherited rotational memory, it is the actively-fed horizon whose two-channel competition forces the phantom-crossing history and, given uneven or rotating accretion, naturally accommodates a directional modulation, both arising from a single feeding process. The axis is permitted by the geometry rather than required by it: isotropic feeding produces the same crossing with no preferred direction, so the anisotropy is a contingent consequence of anisotropic accretion, not a forced prediction. This is a qualitative expectation rather than a derived prediction: because the boundary is a temporal surface (§3.5), anisotropy in the accretion flow enters the interior as a large-scale directional modulation rather than as local structure, which is the correct character for the observed low-multipole anomalies. Whether the amplitude matches requires the junction analysis of §6 and is unexplored here.

The efficiency deficit. The fitted $\kappa \simeq 0.85$ leaves $\sim 15\%$ of the infalling rest energy unaccounted for by the vacuum channel. The constraint is loose ($\kappa = 0.5$ costs only $\Delta\chi^2 \simeq 1.6$, $\sim 1.3\sigma$), but the deficit has natural homes. Standard accretion physics radiates 6–42% of rest energy in the parent's frame before matter crosses the horizon [47–49]; a 15% loss corresponds, under thin-disc assumptions, to a briskly spinning parent ($a_* \sim 0.9$) [50] — the same property that would imprint the preferred axis above, making the deficit and the axis potential fingerprints of a single parameter. The DR2 fit's larger deficit ($\sim 31\%$, at $\kappa \simeq 0.69$) sits more squarely within this 6–42% radiative band — at a correspondingly higher spin — so this reading is robust to, indeed eased by, the shift; only the illustrative a_* value moves. Alternatively, for a rotating hole the swallowed energy genuinely partitions between irreducible (area) growth and rotational energy, so a share retained by the boundary is not excluded; and a small fraction may arrive as interior radiation rather than vacuum. The fitted model is agnostic among these homes: $1 - \kappa$ is a catch-all for energy that does not become interior vacuum.

Implications if correct. Dark energy would be the vacuum energy of the region we inhabit, its magnitude set by the size of the object we live inside, its evolution the record of that object's accretion; the Big Bang a bounce; the cosmological "constant" not constant; the coincidence problem a statement about the parent's feeding history; the universe possessed of a boundary, a parent, and plausibly a lineage. The scale of these implications bears on the model's interest, not on its probability; the probability rests on Sections 4 and 6.

9 Conclusion and decisive tests

Two channels forced by one one-way feeding — energy delivered inward, a boundary pushed outward — generate a dark energy that rises, peaks, and declines, crossing $w = -1$ without phantom matter, in an order the model cannot reverse. Fit to Planck, DESI, and the full Pantheon+ sample, the mechanism matches the empirical description of the evolving-dark-energy signal to within rounding, places the crossing and the peak where the data put them, and selects a conversion efficiency inside the physical budget. A six-model Bayesian comparison on the same vector favours it over the w_0–w_a, constant-w, and running-vacuum alternatives ($\Delta \ln \mathcal{Z} \simeq +2.5$ to $+2.7$) and leaves it statistically indistinguishable from ΛCDM and thawing quintessence, disfavoured by none (Appendix D) — reproducing the empirical preference from a physical origin while paying no evidence penalty for it; on the newer DESI DR2 vector the fit strengthens ($\Delta\chi^2 \simeq 10$) and *the mechanism becomes the evidence-preferred of the six* (Appendix E). The dilution channel is the Misner–Sharp identity; the injection accounting follows the exact Oppenheimer–Snyder matching; the clock ambiguity is absorbed by the fitted history.

Two tests decide it. Theoretically: the junction analysis for an actively fed (Vaidya-type) boundary (Appendices A–C), on which four separate questions now converge — the interior-volume accounting that generates the crossing, the partition of swallowed energy between interior and boundary, whether the large-scale cutoff of Section 8 imprints at all, and the profile and normalisation of the flux whose field-theoretic realisation (Section 3.5) remains open. Empirically: the first-pass Boltzmann implementation reported in Section 4 (tabulated $w_{\text{eff}}(a)$ through PPF, perturbations stable across the crossing, pipeline validated against CAMB to 0.01%) should be extended to a full likelihood analysis over the complete Planck, DESI, and supernova datasets with the interaction's perturbation sector derived rather than assumed minimal — and beyond that, the evolving-dark-energy signal itself, strengthened across DESI's releases from $\sim 2\sigma$ to 2.8–4.2σ [2], will either mature into a measurement or dissolve. The model makes one bet — that the signal is real and has the two-channel shape — and it is stated precisely enough to lose.

Acknowledgements

The hypothesis, conceptual development, research direction, and all analytical and interpretive decisions are the author's own. The author directed the work throughout — identifying the problems, driving the successive refinements of the model, and adjudicating every result. Claude (Anthropic) served as an instrument of that direction, formalising the mathematics, performing the computations, and drafting the text, and equally as an adversary, pressuring the author's proposals on physical and statistical grounds so that ideas were retained only after surviving challenge; several key turns in the model's development followed from the author overriding or correcting the tool. The general-relativistic scoping of Appendices A–C, and an independent reproduction of the numerical pipeline as a validation check, were developed through the same directed, adversarial dialogue with ChatGPT (OpenAI). All scientific interpretations and conclusions remain the author's sole responsibility.

Data and code availability. The analysis pipeline, reproduction scripts, and this manuscript's source are available at `https://github.com/ChrisPGreen/two-channel-dark-energy`, archived at Zenodo, `doi:10.5281/zenodo.21206147` (concept DOI — resolves to the latest version).

The supernova data are the public Pantheon+ release; DESI and Planck inputs are as cited.

Appendices

Supporting derivations for the two-channel mechanism.

A The injection channel from the Vaidya flux

The ingoing Vaidya metric, $ds^2 = -\big(1 - 2GM(v)/c^2r\big)\,dv^2 + 2\,dv\,dr + r^2 d\Omega^2$, carries the stress tensor $T_{ab} = \big[\dot{M}(v)/4\pi r^2\big]\,k_a k_b$ with k_a null [18]. The total energy crossing any sphere during dv is $\oint T_{ab}k^a k^b\,dA\,dv = dM$ *exactly*: the r^{-2} of the flux cancels the sphere area, so the transfer is a global increment with no preferred angle and no locally defined "amount per place." The physically available source strength is therefore set by the growth *rate* $\dot{M}_p$, not by the accumulated mass. Eq. (3) already respects this: by Eq. (2), $\mu_0 a^n\, M_p(a) = dM_p/d\ln a$, so the injection term is $\kappa\,(dM_p/d\ln a)/a^3$ — the rate law in volume-distributed form. Distributing dM_p over the interior physical volume $V \propto a^3$ then yields the a^{-3} scaling; no alternative exponent arises without relocating where the injected energy resides (Appendix C).

B The dilution channel from horizon slaving

The Misner–Sharp mass [28] of a spherically symmetric spacetime is $M_{\rm MS} = (c^2R/2G)\big(1 - g^{ab}\nabla_a R\,\nabla_b R\big)$, and an apparent horizon sits at $R_A = 2GM_{\rm MS}/c^2$. For an interior whose vacuum density is set by the horizon condition, $M = \frac{4\pi}{3}\rho R_A^3$ fixes $\rho = 3c^6/32\pi G^3 M^2$, whence $d\ln\rho/d\ln M = -2$ identically: $\dot{\rho}_{\rm dil} = -2\,(\dot{M}_p/M_p)\,\rho$. Two properties matter. The coefficient is geometric, not fitted; and the channel is *gated by feeding* — if accretion stops, $\dot{M}_p = 0$ and dilution ceases — distinguishing it from ordinary FRW volume dilution ($-3H\rho$), which plays no role for this horizon-slaved component. In spherical symmetry the Vaidya mass function coincides with the Misner–Sharp mass, so a single M_p legitimately drives both this channel and that of Appendix A.

C Scope of the junction analysis and the rapid-conversion limit

Three layers. The parent's accretion connects to Eq. (3) through three layers. (i) Parent-spacetime accretion (Appendix A) fixes what crosses the boundary: $dE = dM_p$. (ii) Junction matching [30, 31] fixes the surface stress-energy induced on the null boundary — an intrinsically two-dimensional quantity. (iii) Conversion and redistribution turn that boundary energy into a smooth interior source; this is transport physics beyond junction theory, and it is the open conversion vertex of §3.5. Layers (i) and (ii) rest on established constructions — the Vaidya accretion of Appendix A and the Barrabès–Israel junction formalism respectively — so that only layer (iii) is genuinely open.

Symmetry does not settle it. Spherical symmetry does not imply homogeneity — a thin shell $\rho \propto \delta\big(r - r_s(t)\big)$ is perfectly isotropic yet maximally inhomogeneous — so matching conditions alone determine what reaches the boundary, not how the interior fills.

The rapid-conversion limit. The minimal completion is a reservoir, $\dot\rho_X = Q - \Gamma\rho_X$, $\dot\rho_{\rm de} = \Gamma\rho_X - D$, with Γ^{-1} the conversion time; the present paper is the rapid-conversion limit $\Gamma \gg H$. A finite Γ introduces a memory kernel whose smearing is largely absorbed into the fitted history (μ_0, n) at background level, but a delay comparable to H^{-1} would desynchronise the channels — dilution responds to horizon growth immediately, injection to past accretion — and could modify the strict form of the ordering theorem (§3.4); that the data are captured by a two-parameter history with no memory freedom is circumstantial evidence against a large delay. The motivation for $\Gamma \gg H$ is the model's own ontology (§3.5): the dark component is horizon-coupled rather than transported, and flux across a boundary whose causal role is temporal enters distributed in space and localised in time.

D Bayesian model selection (DESI DR1)

Information criteria penalise parameter count but not prior volume; a fuller test is the Bayesian evidence $\mathcal{Z} = \int \mathcal{L}(\theta)\,\pi(\theta)\,d\theta$, which integrates the likelihood over the entire prior and so penalises a model for prior volume that does not yield good fits. We computed $\ln\mathcal{Z}$ by nested sampling [51, 52] for six models on the full DR1 vector, under the physical efficiency prior $\kappa \in [0, 1]$: ΛCDM, constant-w (wCDM), running-vacuum (RVM), thawing quintessence, the empirical w_0–w_a form, and the two-channel mechanism (Table 2). The mechanism is evidentially *indistinguishable* from ΛCDM ($\Delta\ln\mathcal{Z} = +0.02$) and from thawing quintessence ($\Delta\ln\mathcal{Z} = -0.10$), the differences lying within nested-sampling scatter, and is *positively-to-strongly favoured* over the dynamical rivals: over the empirical w_0–w_a form by $\Delta\ln\mathcal{Z} \simeq +2.5$ (odds $\simeq 12:1$), and over wCDM and RVM by $\Delta\ln\mathcal{Z} \simeq +2.7$ (odds $\simeq 15:1$). That the mechanism is favoured over w_0–w_a *despite* carrying one more parameter is the substantive result: it reproduces the same fit while concentrating its prior predictions, whereas the empirical form pays a larger volume penalty for the region of (w_0, w_a) space its wide prior never uses.

Two disciplines attach to this result. First, the two-channel-versus-w_0–w_a comparison was cross-validated by two independent codepaths — a dedicated single-comparison run ($\Delta\ln\mathcal{Z} = +2.69$) and the six-model suite ($+2.50$) — agreeing within their combined error, for a pooled estimate $\Delta\ln\mathcal{Z} \simeq +2.6 \pm 0.4$; the RVM evidence is prior-sensitive and is reported with its prior width documented in the accompanying validation package. Second, a structural contrast sharpens the ties: thawing quintessence is confined to $w \geq -1$ and structurally *cannot* cross the phantom divide (Figure 2), so its evidential tie with the mechanism holds only while the data do not demand a crossing — a model that forces the crossing and one that forbids it are separated not by present evidence but by exactly the expansion-history measurement of Section 9.

The advantage is tied to the physical bound. That the mechanism is favoured over w_0–w_a is a property of the *physical* prior, not of the extra parameters as such. Because the likelihood is negligible for $\kappa > 1$ (the fit selects $\kappa \simeq 0.85$), widening the efficiency prior from its physical range $\kappa \in [0, 1]$ to the frequentist stress range $\kappa \in [0, 40]$ merely dilutes it over unused, physically impossible efficiencies: the evidence then falls by the prior-volume ratio, $\Delta \ln \mathcal{Z} \simeq \ln 40 \simeq 3.7$. The mechanism's Bayesian edge is therefore an accounting of a genuine physical constraint — $\kappa \leq 1$ is a law, not a tunable prior width — and it evaporates exactly when that constraint is artificially removed.

Table 2: Bayesian evidence for six models on the full DR1 vector (Planck distance priors, DESI DR1 BAO, and the complete Pantheon+ sample), by nested sampling under the physical efficiency prior $\kappa \in [0, 1]$. k counts free parameters; $\ln \mathcal{Z}$ is the log-evidence (higher is better, Occam-penalised for prior volume); $\Delta \ln \mathcal{Z}$ is measured relative to the two-channel mechanism, with + favouring it. χ^2_{min} is the best point located by the sampler — an evidence-run by-product marginally above the dedicated optima quoted in Table 1. Jeffreys scale on $|\Delta \ln \mathcal{Z}|$: < 1 inconclusive, 1–2.5 positive, 2.5–5 strong.

Model	k	χ^2_{min}	$\ln \mathcal{Z}$	$\Delta \ln \mathcal{Z}$ vs. two-channel
Thawing quintessence	3	1406.8	-712.43 ± 0.23	-0.10 (tie)
Two-channel mechanism	5	1403.4	$\mathbf{-712.53 \pm 0.24}$	—
ΛCDM	2	1409.4	-712.55 ± 0.22	$+0.02$ (tie)
w_0–w_a (CPL)	4	1403.5	-715.03 ± 0.28	$+2.50$ (positive)
wCDM (constant w)	3	1408.0	-715.22 ± 0.26	$+2.69$ (strong)
RVM (running vacuum)	3	1407.8	-715.24 ± 0.26	$+2.71$ (strong)

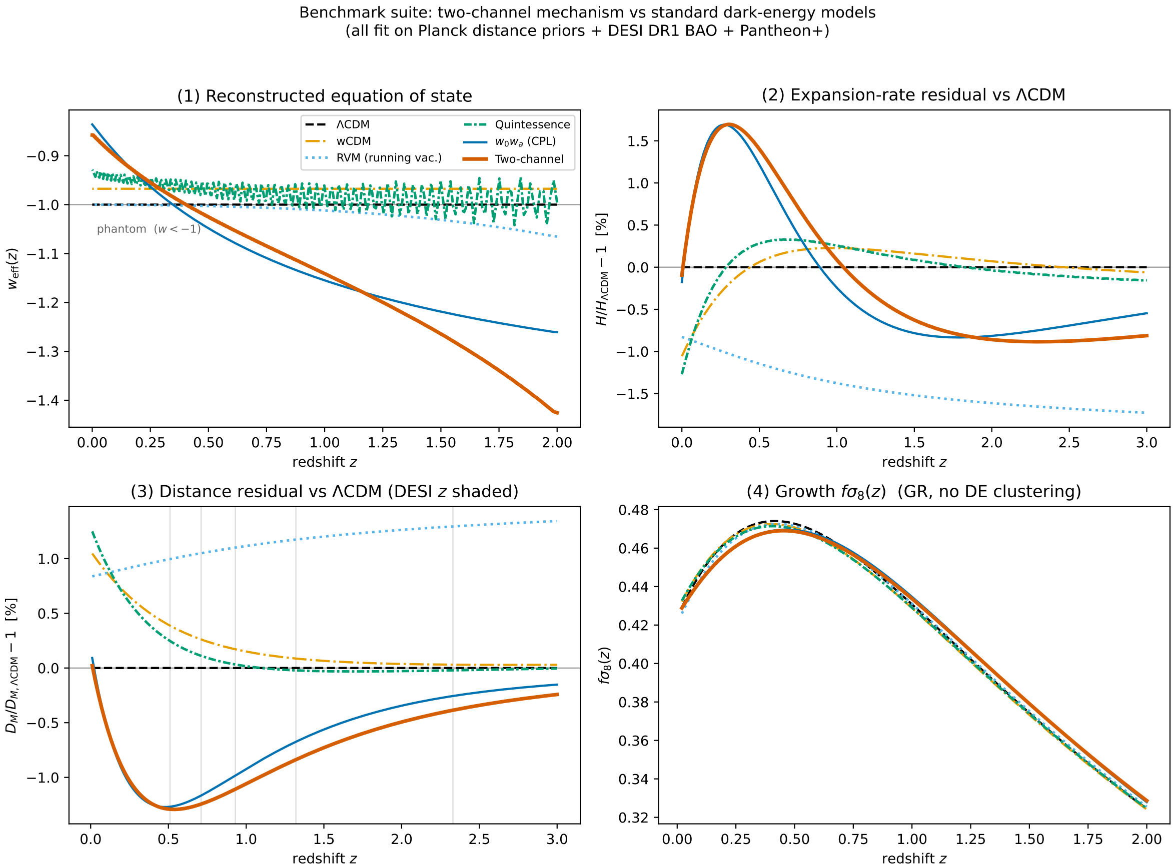

Figure 2: Model comparison on the fitted DR1 vector (Planck distance priors, DESI DR1 BAO, and the complete Pantheon+ sample). *Top left:* reconstructed effective equation of state $w_{\rm eff}(z)$ — ΛCDM flat at -1, thawing quintessence pinned at $w \geq -1$ (it structurally cannot cross the phantom divide), and both the empirical w_0–w_a form and the two-channel mechanism crossing $w = -1$ from below near $z \simeq 0.4$. The two-channel curve then continues steeply downward at $z \gtrsim 1$ (to $w_{\rm eff} \simeq -1.4$ by $z = 2$); there the dark-energy density is small and matter-dominated, so this is the equation of state of a dynamically subdominant component and does not distort observables. *Top right and bottom left:* fractional $H(z)$ and $D_M(z)$ residuals against ΛCDM — sub-percent for the mechanism, which tracks w_0–w_a, while running-vacuum drifts oppositely. *Bottom right:* the growth observable $f\sigma_8(z)$, on which all six models coincide to within 2% (no dark-energy clustering). Produced by `benchmark_plots.py` in the validation package.

E Against DESI DR2 data

The main text fits the DESI first-release (DR1) BAO vector throughout (§4). As a robustness check we repeat the full analysis — fits and six-model evidence — against the DESI second-release (DR2) BAO [2], holding the Planck distance priors and the complete Pantheon+ sample fixed. Only the BAO block changes: from the five DR1 anisotropic tracers to the six DR2 tracers (LRG, ELG, QSO, Lyα; the isotropic BGS point is omitted, in parallel with the DR1 treatment). Nothing in the mechanism depends on the release — the master equation (3), the rise–peak–decline structure, and the forced ordering of §3.4 are unchanged; only the fitted history and its confrontation with data move.

Fit. On the DR2 vector the best-fit values are $\chi^2 = 1408.2$ (ΛCDM), 1399.2 (w_0–w_a), and 1398.2 (two-channel). The evolving-dark-energy preference over ΛCDM is $\Delta\chi^2 \simeq 10$ for three extra parameters — up from $\simeq 6$ on DR1 — and the mechanism now marginally *edges* the empirical w_0–w_a form on fit ($\Delta\chi^2 \simeq 1.0$) rather than tying it. The reconstruction remains physical: the $w = -1$ crossing and the density peak move to $z \simeq 0.51$, and the fitted efficiency *falls* from $\kappa \simeq 0.85$ on DR1 to $\kappa \simeq 0.69$ on DR2 — both comfortably inside the budget $\kappa \leq 1$, so the tighter data continue to select a physical conversion efficiency, unforced and now further from the ceiling. The tighter DR2 constraint therefore reinforces the same shape at a physical efficiency (Figure 3).

Evidence. Repeating the six-model nested-sampling comparison on the DR2 vector under identical priors (Table 3) sharpens the DR1 result. The two-channel mechanism is now the *evidence-preferred* model of all six ($\ln\mathcal{Z} = -711.34$): *strongly* favoured over constant-w ($\Delta\ln\mathcal{Z} = +3.97$) and running-vacuum (+2.55), *positively* favoured over the empirical w_0–w_a form (+2.30), and nominally favoured over ΛCDM (+1.02); only thawing quintessence remains evidentially *indistinguishable* (+0.12). Where DR1 left the mechanism level with ΛCDM and a hair behind quintessence, DR2 places it best of the six — a step above ΛCDM and level with quintessence.[1]

As the DESI constraint tightens from DR1 to DR2, the mechanism moves from tied-best to nominally best on the evidence while its shape and physical efficiency are preserved — the behaviour expected if the evolving-dark-energy signal is real and has this origin, and the opposite of what a two-parameter artifact would do.

Table 3: Bayesian evidence for the same six models on the DESI *DR2* vector (Planck distance priors, DESI DR2 BAO [2], and the complete Pantheon+ sample), by nested sampling under the physical efficiency prior $\kappa \in [0, 1]$. Columns as in Table 2; $\Delta\ln\mathcal{Z}$ is relative to the two-channel mechanism, with + favouring it.

Model	k	$\chi^2_{\min}$	$\ln\mathcal{Z}$	$\Delta\ln\mathcal{Z}$ vs. two-channel	
Two-channel mechanism	5	1398.5	$\mathbf{-711.34 \pm 0.25}$	—	
Thawing quintessence	3	1404.5	-711.46 ± 0.23	+0.12	(tie)
ΛCDM	2	1408.2	-712.37 ± 0.22	+1.02	(positive)
w_0–w_a (CPL)	4	1399.3	-713.64 ± 0.29	+2.30	(positive)
RVM (running vacuum)	3	1404.4	-713.89 ± 0.27	+2.55	(strong)
wCDM (constant w)	3	1406.6	-715.31 ± 0.27	+3.97	(strong)

[1] On the DR2 vector the two-channel posterior is narrow enough that the default nested-sampling proposal stalls; the DR2 evidences use slice sampling (`rslice`) uniformly across the six models for numerical robustness, which does not bias the estimated evidence. The DR1 evidences of Table 2 are unchanged.

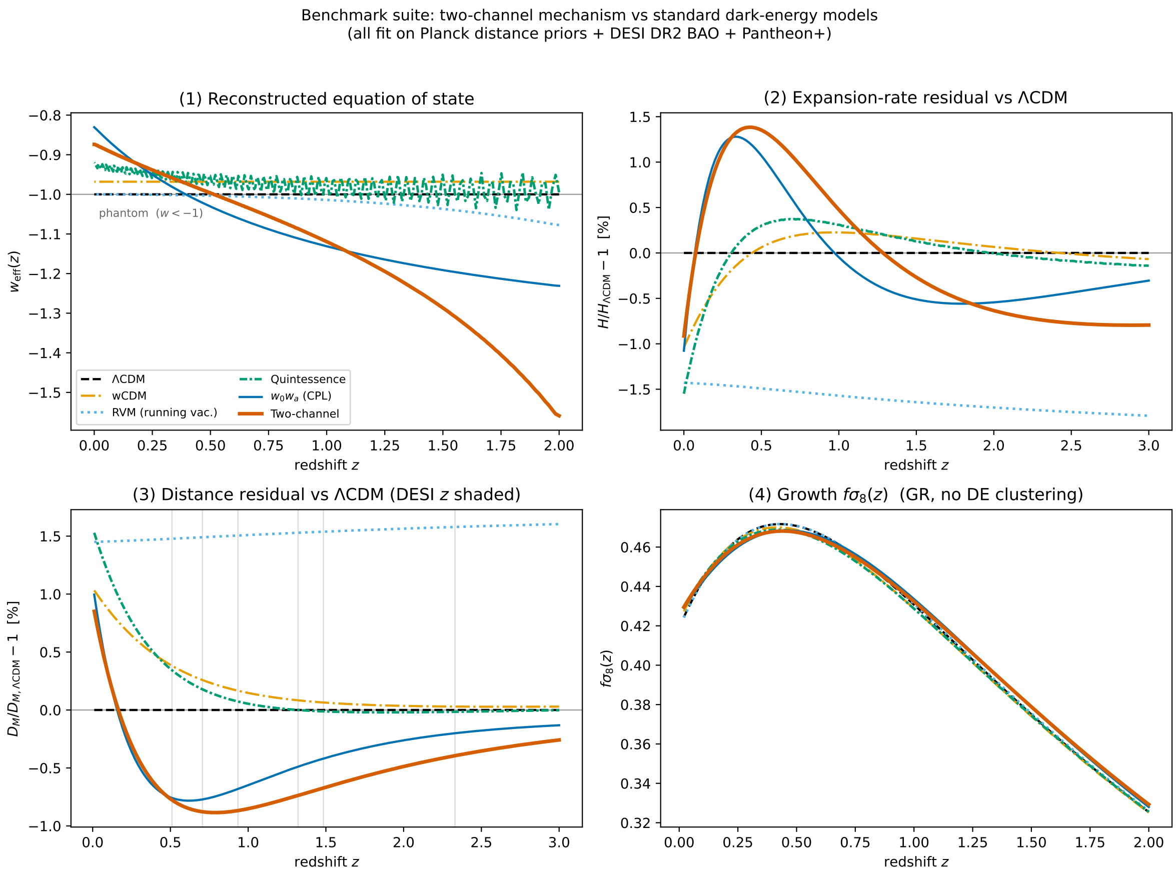

Figure 3: Model comparison on the DESI *DR2* vector (cf. Figure 2 for DR1). *Top left:* reconstructed $w_{\rm eff}(z)$ — the two-channel mechanism and w_0–w_a cross $w=-1$ from below (now near $z \simeq 0.51$), while thawing quintessence stays $w \geq -1$. *Top right and bottom left:* fractional $H(z)$ and $D_M(z)$ residuals against ΛCDM, with the DR2 BAO redshifts shaded. *Bottom right:* the growth observable $f\sigma_8(z)$, on which all six models coincide to within 2%. Produced by `benchmark_plots.py --dr2`.

Version note

Version 1.1 (July 2026). This revision adds a six-model Bayesian evidence comparison (Appendix D, Table 2, Figure 2), a confrontation with the DESI second data release (Appendix E, Table 3, Figure 3), and a literature-novelty analysis (Section 7.2), and states the DR1/DR2 data provenance explicitly. The main-text argument — the mechanism, the frequentist fit of Table 1, and the forced-ordering theorem — is unchanged from v1.0.2; the conservative two-channel cell is corrected from 13.9 to its reproducible physical-ceiling value 13.8 (at $\kappa = 1$), a shift of 0.1 that changes no conclusion. This manuscript version numbers the paper only; the reproduction package is versioned independently in the repository, so the software can be revised without reissuing the paper. Both stand at v1.1 for the present release. A persistent record of all versions is archived at the concept DOI on the title page.

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